

To install KNIME:

- Download the KNIME Desktop Application from <https://www.knime.com/>.
- Install this application and provide the necessary permissions during the setup process.

Decision tree

A decision tree is a decision support tool that uses a tree-like model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility. It is one way to construct a model that contains conditional control statements. Decision trees are used for handling non-linear data sets effectively. The decision tree tool is used in real life in many areas, such as engineering, civil planning, law, and business.

Designing a supervised ML car evaluation decision tree model

First, create a new workflow for the car evaluation model. Drag the .csv file into KNIME to create and initialise the *CSV Reader* node. Connect it to a colour manager, where each genre can be assigned a specific colour under ‘swatches’ or a preset under ‘palettes’.

Next, connect the colour manager to the *partitioning* node. This is used to separate the given data set into two partitions -- one consisting of 80 per cent of the data and the other consisting of the remaining 20 per cent of the data. Thus the data is partitioned into two sets for testing and training in an 80:20 ratio. This partition is done by choosing the ‘randomly’ option.

Now we need to connect the *partitioning* node to the decision tree learner. Configure it as a class column as units, and change the records as per the required analysis. Next, connect the decision tree learner to the decision tree predictor. Send the smaller partition (test data) to the decision tree learner in order to classify the values, which can then be predicted by the decision tree predictor.

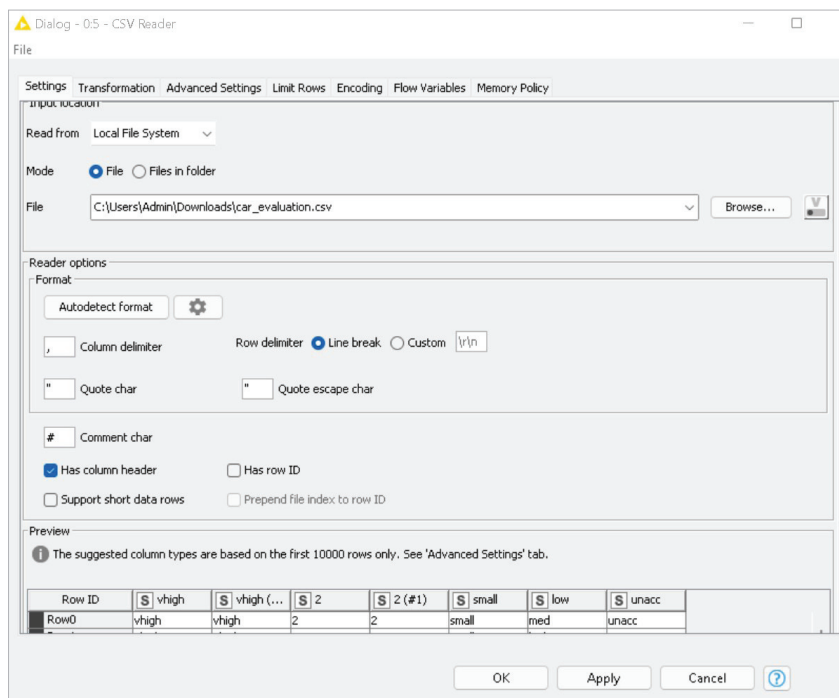


Figure 2: Configuration of *CSV reader* node

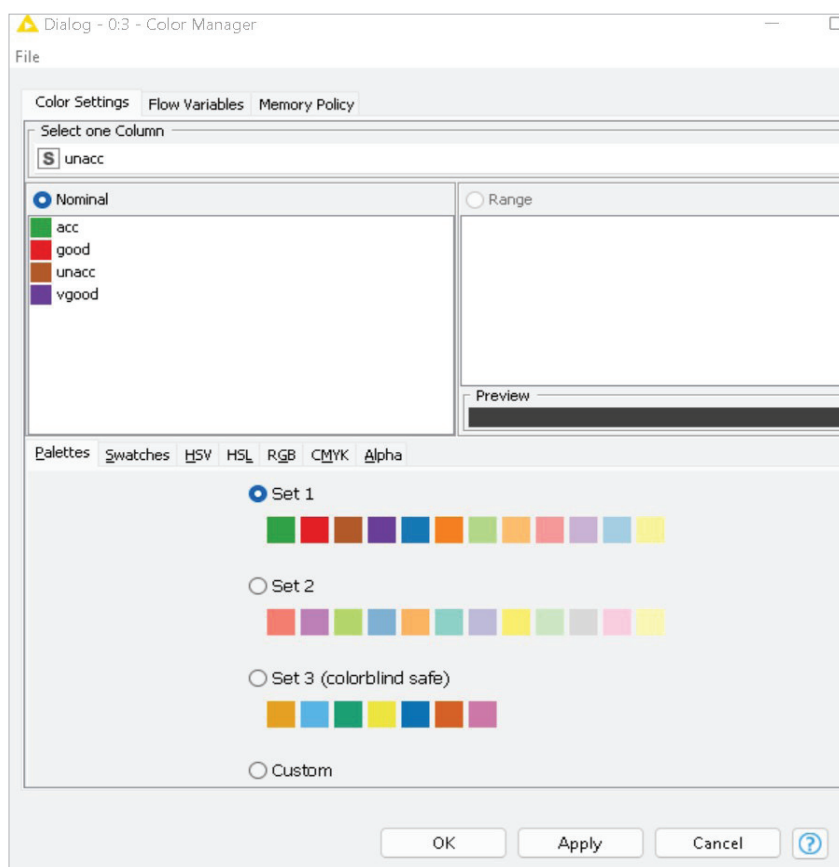


Figure 3: Configuration of *colour manager* node